

# VED NAIK

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## EDUCATION

**The Pennsylvania State University, University Park**

**Aug 2024 – May 2026 (Expected)**

*GPA: 3.89 | M.S. in Computer Science and Engineering*

*State College, PA*

*Assistantships: TA for CMPSC 132 - Programming and Computation II: Data Structures*

**NMIMS University**

**Aug 2020 – May 2024**

*CGPA: 3.83 | B.Tech. in Computer Engineering, Minors in Machine Learning and Data Analytics*

*Mumbai, IND*

## TECHNICAL SKILLS

**Programming Languages:** Python, Java, C++, TypeScript, JavaScript

**Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB

**Dev Tools:** Spring, FastAPI, BeautifulSoup, Angular, RxJS, Node.js, React, Django, Docker, Kubernetes, CI/CD, Git, Kafka

**Testing and Cloud:** PyTest, Unittest, Selenium, Jasmine, Karma, AWS

**AI/ML Tools:** PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, LangChain, Ollama, MCP/FastMCP, Open WebUI

## EXPERIENCE

**Gametime Hero, Inc.**

**Jun 2025 – Aug 2025**

*Software Engineer Intern*

*Seattle, WA*

- Built Angular components and services for a sports platform, improving page responsiveness and **cutting load times by 40%**.
- Integrated Firebase Authentication with role-based access control, securing sensitive player data for **4,000+ users**.
- Wrote unit tests using Jasmine and Karma, **increasing frontend code coverage to 80%** and ensuring code reliability.
- Identified and resolved 4 critical Angular bugs, reducing production errors and improving platform stability.

**Techaroha Solutions Private Limited**

**Jan 2024 – Apr 2024**

*Software Engineer Intern*

*Navi Mumbai, IND*

- Engineered a project management system using Express.js/Angular/MongoDB, improving tracking and **asset access by 40%**.
- Containerized development environments with Docker and standardized service configurations, reducing setup time.
- Architected a **3-tier** AWS image recognition pipeline using SQS, S3, and EC2, improving request handling reliability.
- Added SQS-driven auto-scaling using Spring Boot, **cutting image processing latency by 50%** and increasing throughput.

**Network Intelligence Pvt. Ltd.**

**May 2023 – Jul 2023**

*Cybersecurity Assessment Team Intern*

*Mumbai, IND*

- Identified security risks by analyzing headers, ports, anomalous packets with **Wireshark**, informing network improvements.
- Investigated network vulnerabilities using **Wireshark** and **Nmap**, analyzing TCP/IP, UDP, SMTP, and ICMP traffic patterns.

**Career Robin**

**Sep 2021 – Nov 2021**

*Data Science Intern*

*Mumbai, IND*

- Increased data processing speed by 36% by creating a Python **web scraper** for **15 job boards**, enhancing NLP analysis.
- Improved efficiency by building NLP pipelines with **NLTK**, **spaCy**, **Gensim**, and **Pandas** for effective text mining.
- Deployed web scraper on AWS for scalable, automated data extraction, **cutting retrieval time by 40%**.

## PROJECTS

**Parallel Me | Next.js, FastAPI, MongoDB Atlas, Supabase Realtime, Google Gemini API**

**Feb 2025 – Mar 2025**

- Architected emotion-matching engine using **Gemini embeddings** and **MongoDB Atlas Vector Search** (Next.js, FastAPI).
- Implemented real-time anonymous chat via **Supabase WebSockets** with Auth0 auth and AI-generated icebreakers.

**CloudCode | React, React Flow, TypeScript, FastAPI, Terraform, AWS**

**Feb 2025 – Mar 2025**

- Spearheaded an AI-powered platform transforming natural language into **Terraform** via a Multi-Agent System and **FastAPI**.
- Designed a FastAPI service that automates standardized Terraform deployments across multiple cloud providers.
- Modeled a **React** frontend with real-time bi-directional sync between visual diagrams and code.

**Intelligent Human Assistant | Python, Ollama, MCP/FastMCP, Open WebUI, RAG-Anything**

**Sep 2025 – Nov 2025**

- Created an **MCP terminal server** exposing tools to start, run, and terminate a persistent shell session with state preservation.
- Converted natural language to validated shell commands with concise output summaries via an **Ollama-based MCP client**.
- Engineered an **MCP RAG server** to preprocess documents into a persistent vector knowledge base for semantic retrieval.
- Developed an agentic automation system using OCR-based UI detection, coordinate mapping, and sequential MCP workflows.
- Integrated terminal control, RAG pipelines, and UI automation into a multi-agent knowledge navigator in Open WebUI.

**Northropp Grumman Object Detection Hackathon 2024 | Python, CLIP, Flask, REST API**

**Oct 2024 – Nov 2024**

- Won **2nd place** in a hackathon by building a real-time object detection system with YOLO, Grounding DINO, and a UI.

## PUBLICATIONS

- **Culinary Curator: An AI-Powered Recipe Assistant** – Published in IEEE Xplore.
- **Enhancing Toxic Comment Classification: A Deep Learning Approach with Pre-trained Language Models** – Published in IEEE Xplore.